

1. Use the Windows command prompt to run the command `ipconfig /all` on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

```
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . : 
    Description . . . . . : Intel(R) Dual Band Wireless-AC 8265
    Physical Address. . . . . : DC-71-96-1E-2B-20
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv6 Address. . . . . : 2405:201:2033:6025:95b1:f216:5d8c:9059(Preferred)
    Temporary IPv6 Address. . . . . : 2405:201:2033:6025:547f:13f3:e4be:6e6a(Preferred)
    Link-local IPv6 Address . . . . . : fe80::eec0:9248:fb84:eee8%13(Preferred)
    IPv4 Address. . . . . : 192.168.29.56(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Friday, July 31, 2026 6:02:57 PM
    Lease Expires . . . . . : Saturday, August 1, 2026 6:02:57 PM
    Default Gateway . . . . . : fe80::6ec2:42ff:fe23:edc9%13
                                192.168.29.1
    DHCP Server . . . . . : 192.168.29.1
    DHCPv6 IAID . . . . . : 81555862
    DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-0E-42-8D-9C-5A-44-7E-66-A9
    DNS Servers . . . . . : 2405:201:2033:6025:c0a8:1d01
                                192.168.29.1
                                2405:201:2033:6025:c0a8:1d01
    NetBIOS over Tcpip. . . . . : Enabled
```

2. Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to `www.google.com`. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

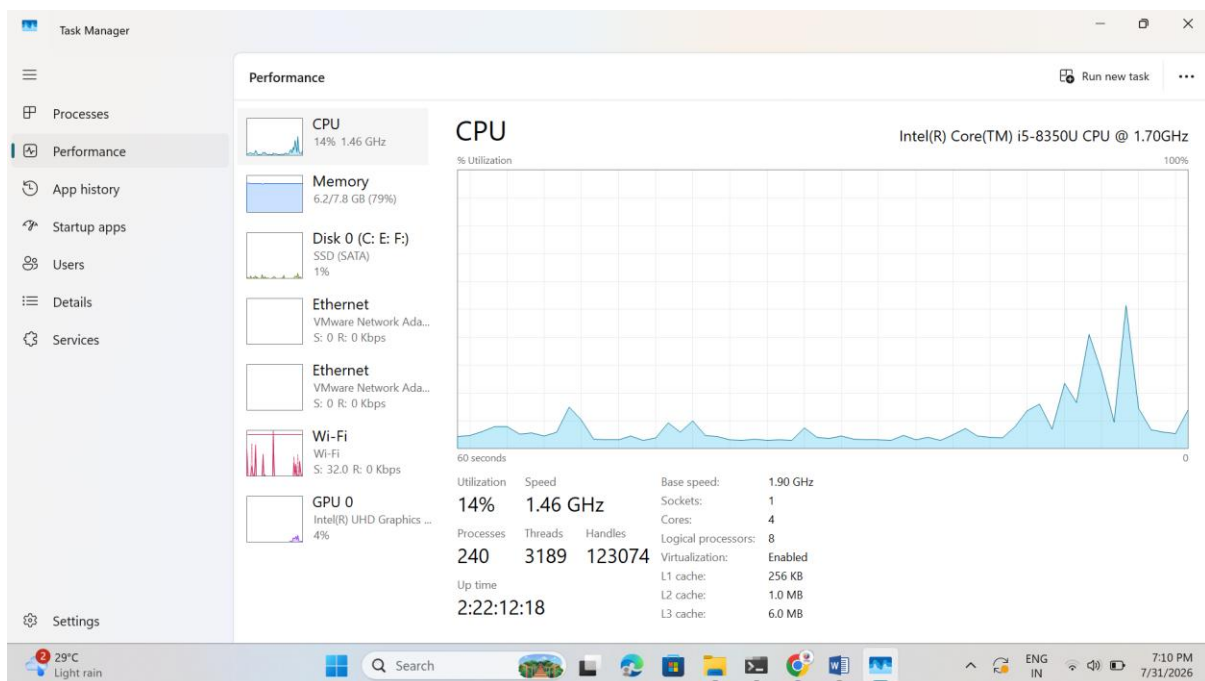
```
C:\Users\hellO>ping www.google.com
Ping request could not find host www.google.com. Please check the name and try again.
```

```
C:\Users\helL0>ping www.google.com

Pinging www.google.com [2001:4860:4829:7700::] with 32 bytes of data:
Reply from 2001:4860:4829:7700::: time=32ms
Reply from 2001:4860:4829:7700::: time=33ms
Reply from 2001:4860:4829:7700::: time=24ms
Reply from 2001:4860:4829:7700::: time=23ms

Ping statistics for 2001:4860:4829:7700:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 23ms, Maximum = 33ms, Average = 28ms
```

3. Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.



**4.Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets

Hint: Use the filter bar in Wireshark and try http or https as the filter keyword**

Information	Description
Source and Destination IP Address	Shows where the packet came from and where it is going
Protocol	Displays the protocol used (HTTP, TLS/HTTPS, TCP, DNS).
Packet Length and Time	Shows packet size and the timestamp of when it was captured.

5.Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

AI-Generated Checklist

1. Restart your Wi-Fi router and modem.
2. Move closer to the Wi-Fi router or use a wired Ethernet connection.
3. Close unnecessary apps or downloads that consume bandwidth.
4. Update the router firmware and network adapter drivers.
5. Use the **5 GHz Wi-Fi band** instead of 2.4 GHz if available.
6. Restart your gaming device before playing.
7. Reduce the number of devices connected to the network during gaming.